CS4048 – DATA SCIENCE

Lecture 27

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FAST – NUCES, CFD Campus

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TODAY'S TOPICS

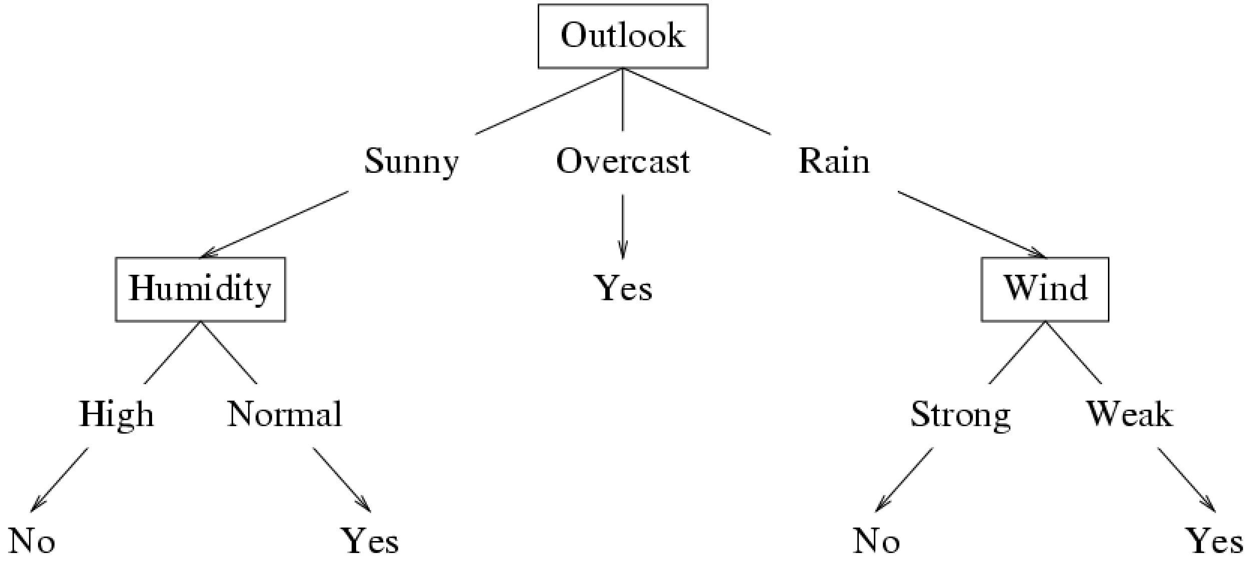
- Decision tree representation
- ID3 learning algorithm
- Entropy, information gain

TRAINING DATA EXAMPLE: GOAL IS TO PREDICT WHEN THIS PLAYER WILL PLAY TENNIS?

Day	Outlook	Temp.	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

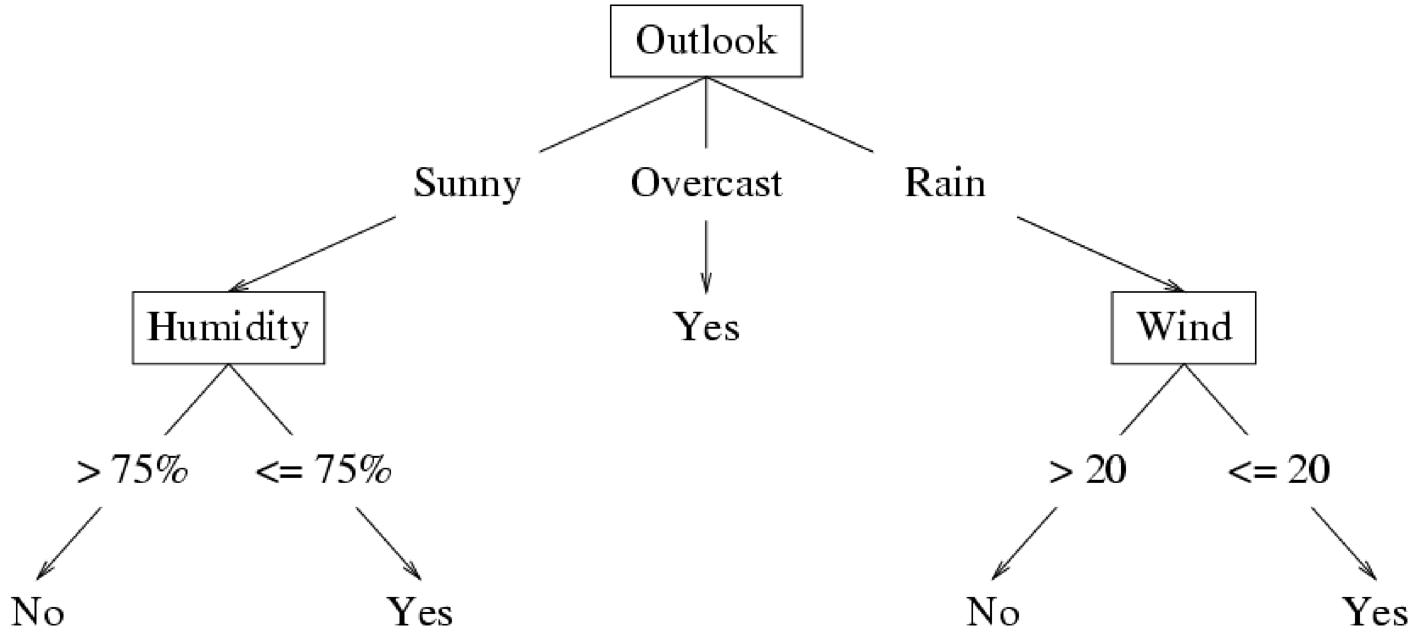
Decision Tree Hypothesis Space

- **Internal nodes** test the value of particular features x_j and branch according to the results of the test.
- **Leaf nodes** specify the class $h(\mathbf{x})$.



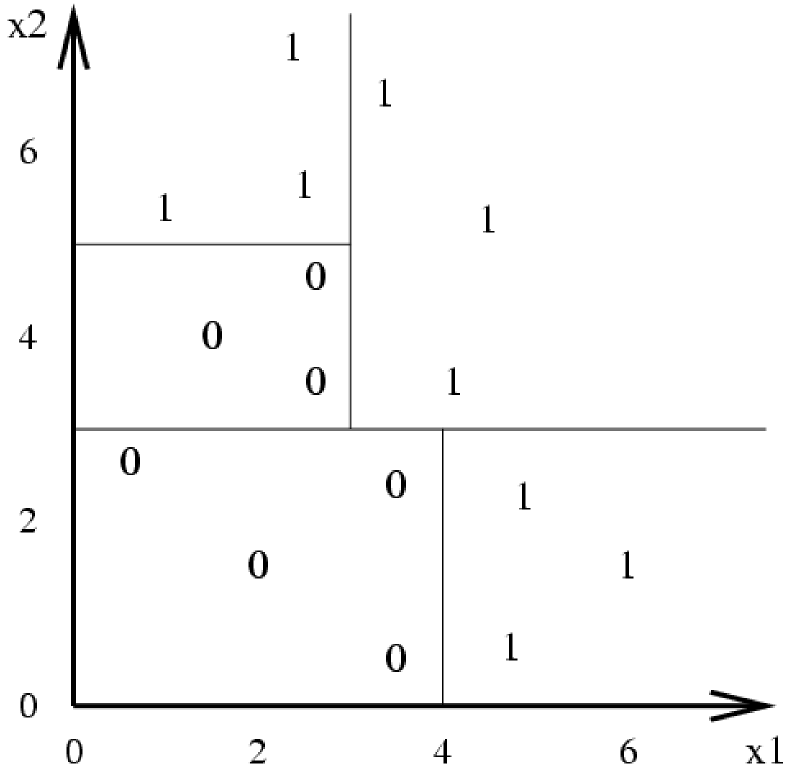
Suppose the features are **Outlook** (x_1), **Temperature** (x_2), **Humidity** (x_3), and **Wind** (x_4). Then the feature vector $\mathbf{x} = (Sunny, Hot, High, Strong)$ will be classified as **No**. The **Temperature** feature is irrelevant.

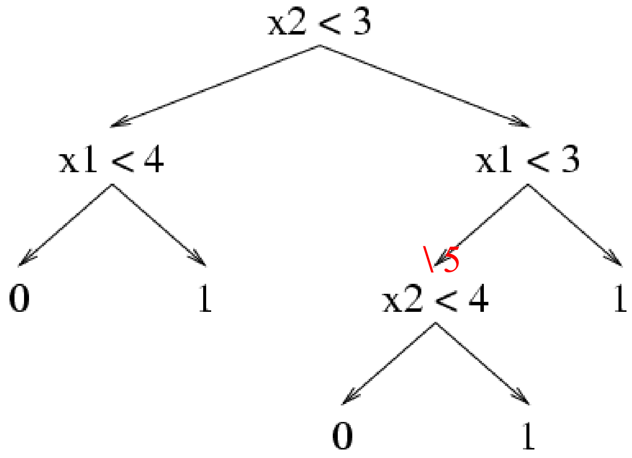
If the features are continuous, internal nodes may test the value of a feature against a threshold.



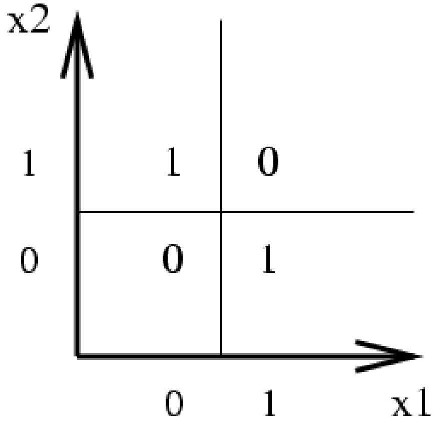
Decision Tree Decision Boundaries

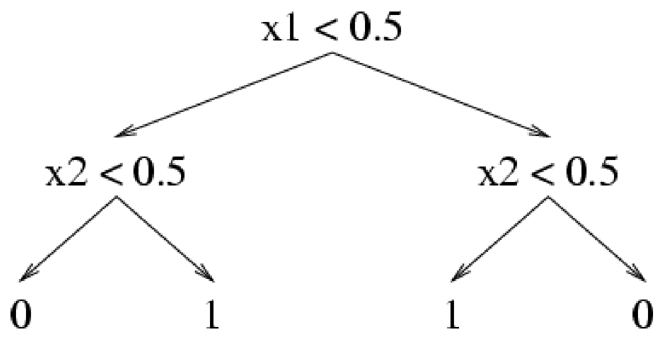
Decision trees divide the feature space into axis-parallel rectangles, and label each rectangle with one of the K classes.





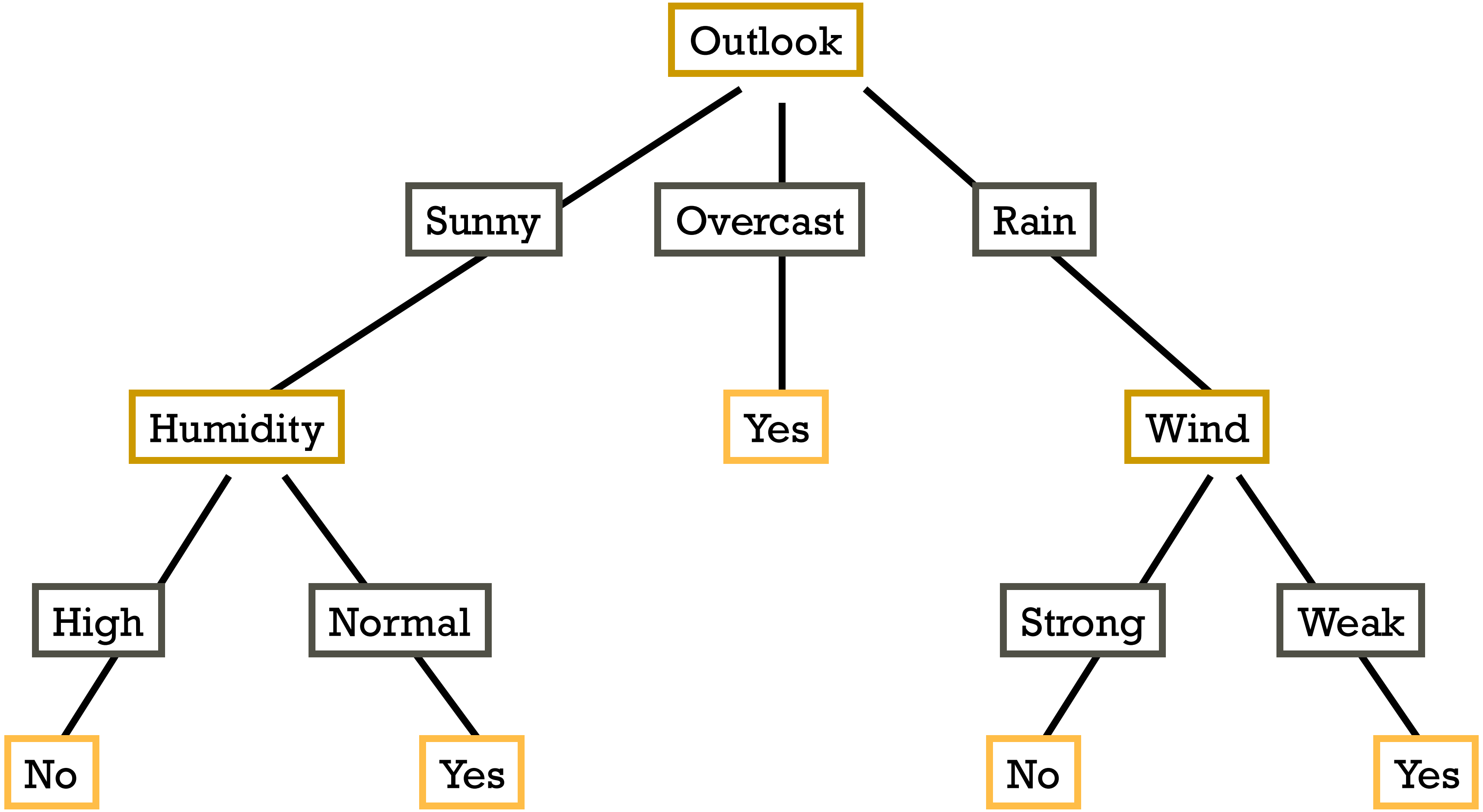
Decision Trees Can Represent Any Boolean Function



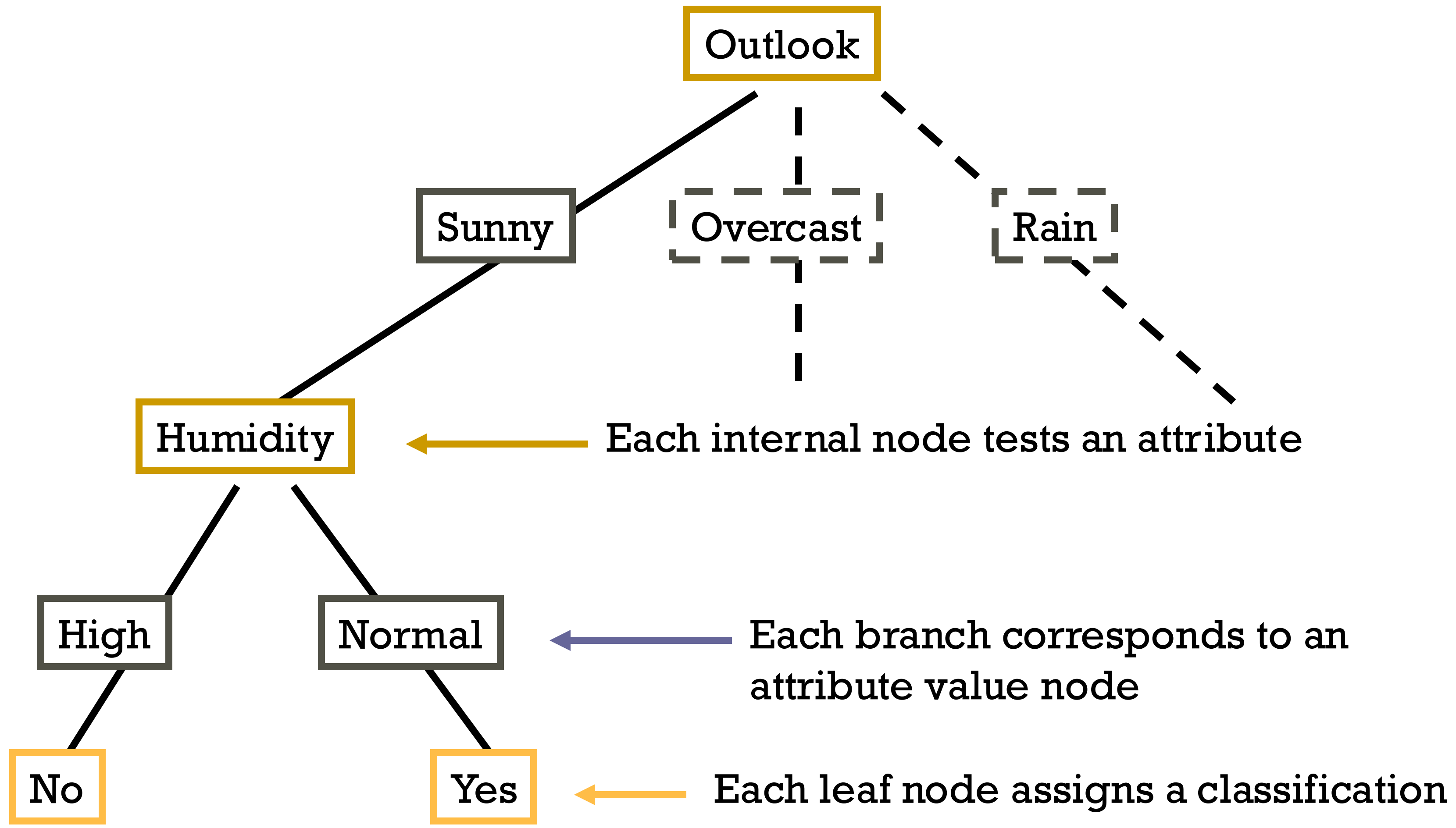


The tree will in the worst case require exponentially many nodes, however.

DECISION TREE FOR PLAYTENNIS



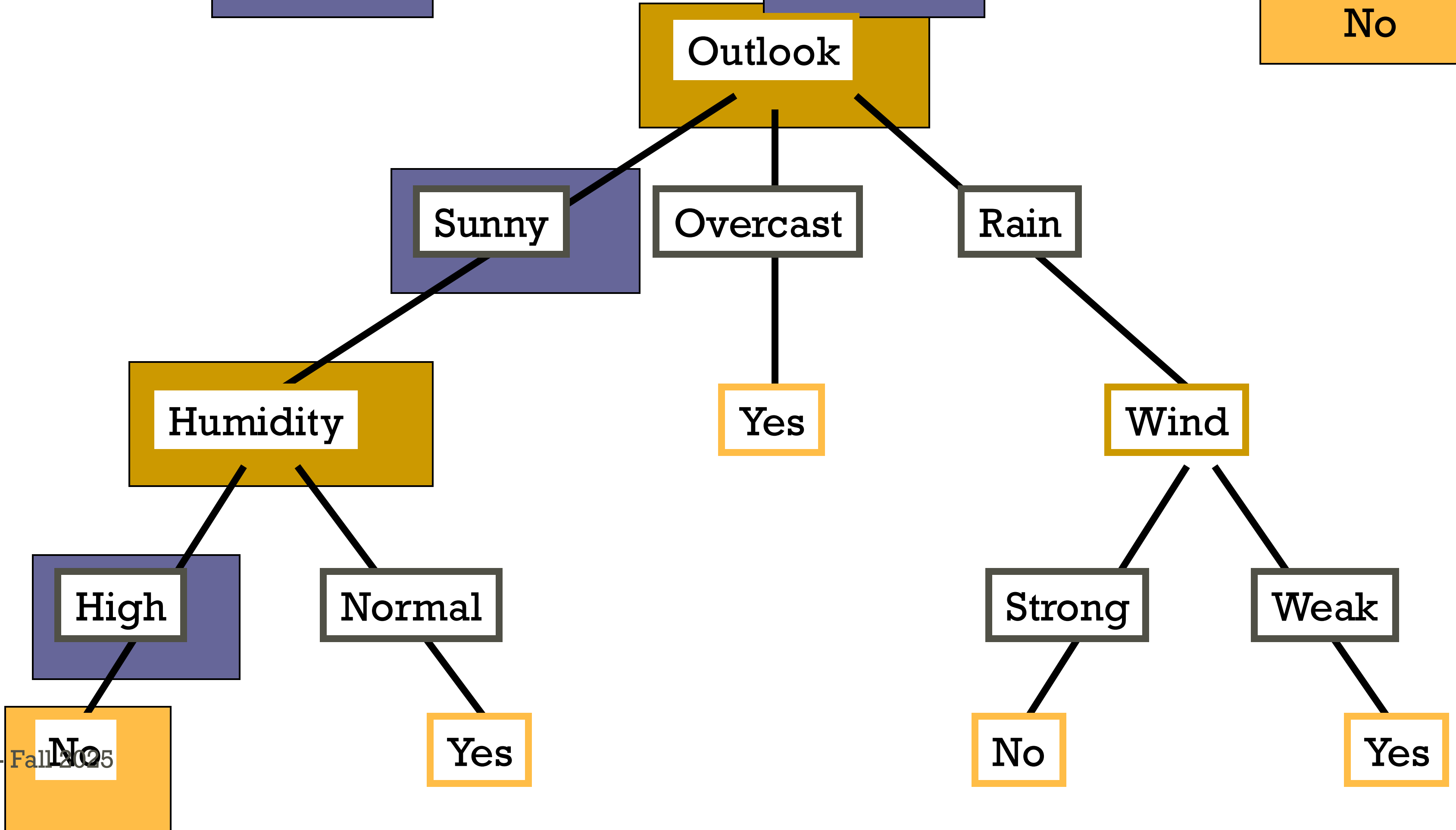
DECISION TREE FOR PLAYTENNIS



DECISION TREE FOR PLAYTENNIS

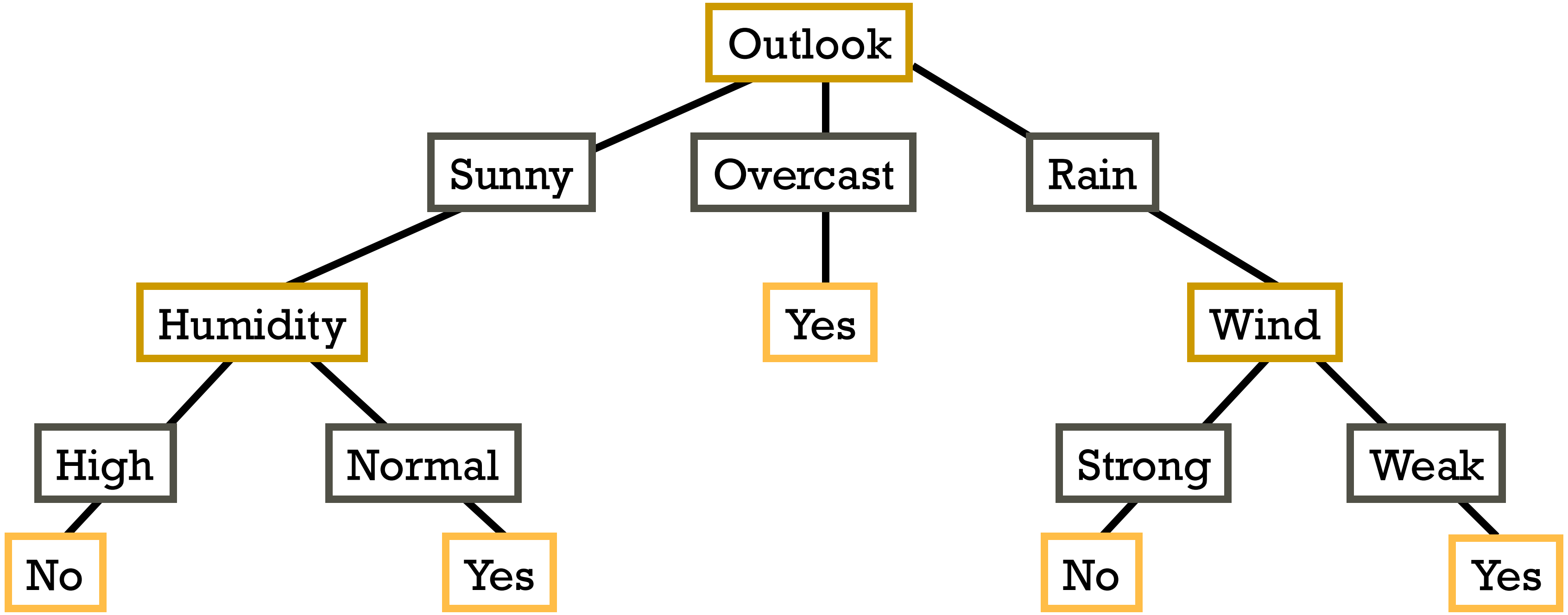
Outlook	Temperature	Humidity	Wind	PlayTennis
Sunny	Hot	High	Weak	?

No



DECISION TREE

Decision trees represent disjunctions of conjunctions



(Outlook=Sunny $\wedge$ Humidity=Normal)
$\vee$ (Outlook=Overcast)
$\vee$ (Outlook=Rain $\wedge$ Wind=Weak)

LEARNING ALGORITHM FOR DECISION TREES

Input dataset: $S = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$

$\textsc{GrowTree}(S)$

if $(y = 0 \text{ for all } \langle \mathbf{x}, y \rangle \in S)$ **return** new leaf(0)

else if $(y = 1 \text{ for all } \langle \mathbf{x}, y \rangle \in S)$ **return** new leaf(1)

else

choose best attribute x_j

$S_0 = \text{all } \langle \mathbf{x}, y \rangle \in S \text{ with } x_j = 0;$

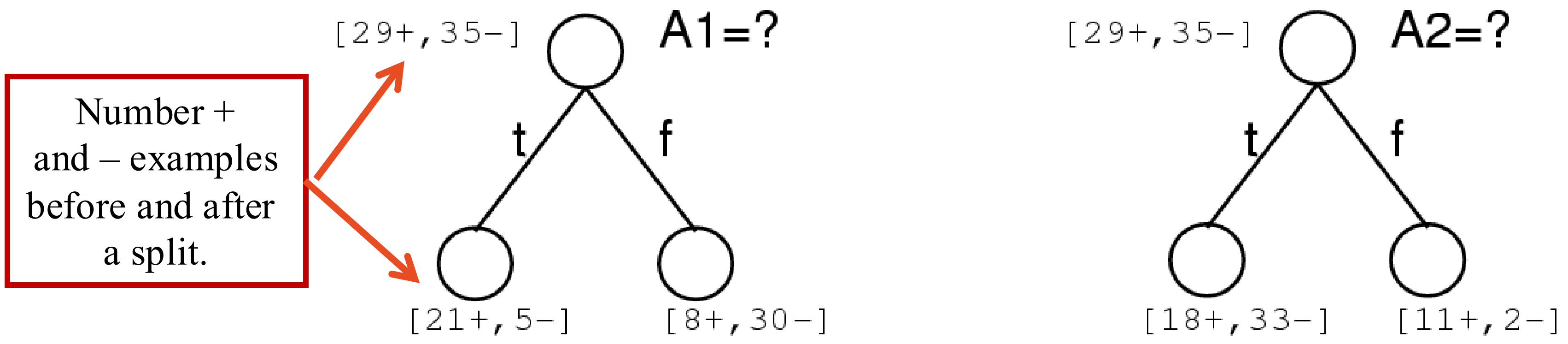
$S_1 = \text{all } \langle \mathbf{x}, y \rangle \in S \text{ with } x_j = 1;$

return new node$(x_j, \textsc{GrowTree}(S_0), \textsc{GrowTree}(S_1))$

CHOOSING THE BEST ATTRIBUTE

A1 and A2 are “attributes” (i.e. features or inputs).

Which attribute is best?

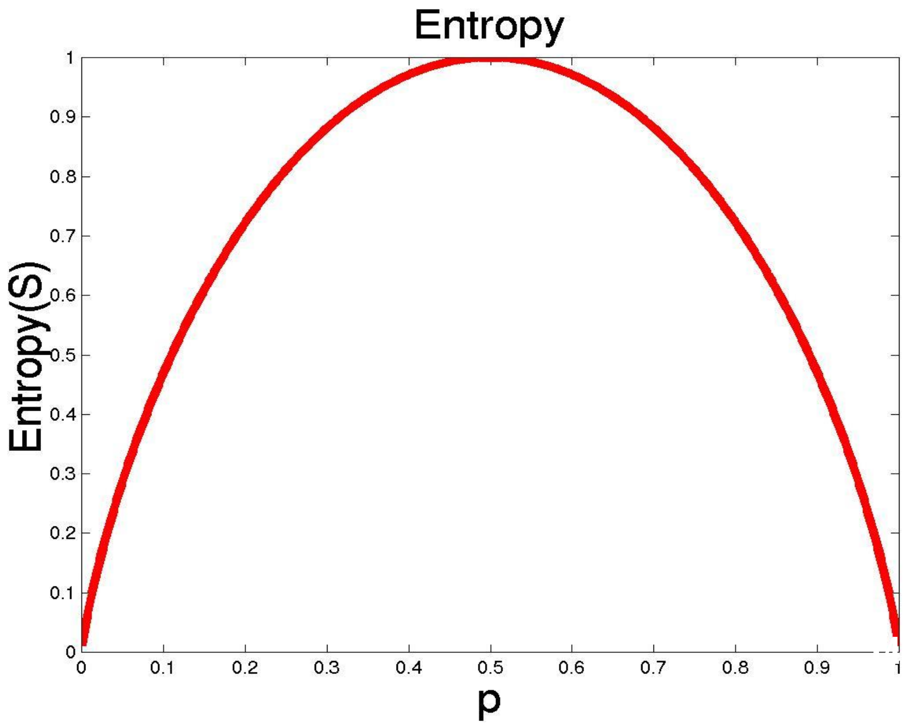


- Many different frameworks for choosing BEST have been proposed!
- We will look at Entropy Gain.

ENTROPY

- S is a sample of training examples
- p_+ is the proportion of positive examples
- p_- is the proportion of negative examples
- Entropy measures the impurity of S

  Entropy(S) = $-p_+ \log_2 p_+ - p_- \log_2 p_-$

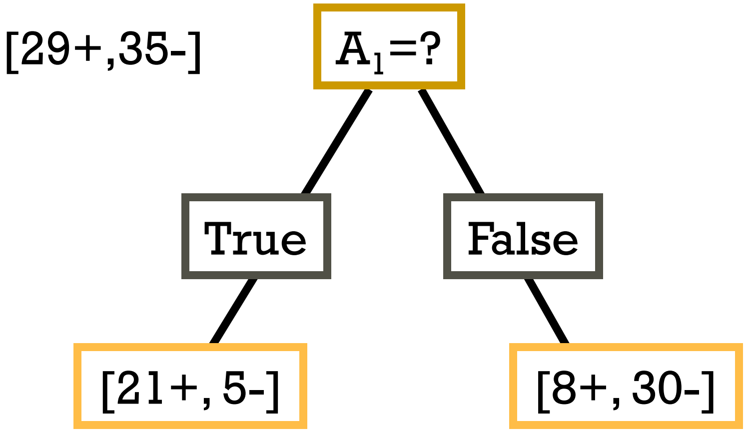


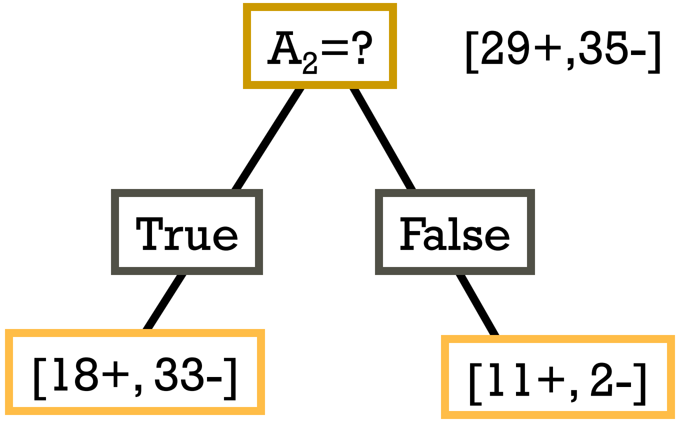
INFORMATION GAIN

- Gain(S,A): expected reduction in entropy due to sorting S on attribute A

$$\text{Gain}(S,A)=\text{Entropy}(S) - \sum_{v\in \text{values}(A)} |S_v|/|S|\ \text{Entropy}(S_v)$$

$$\text{Entropy}([29+,35-]) = -29/64 \log_2 29/64 - 35/64 \log_2 35/64$$
$$= 0.99$$



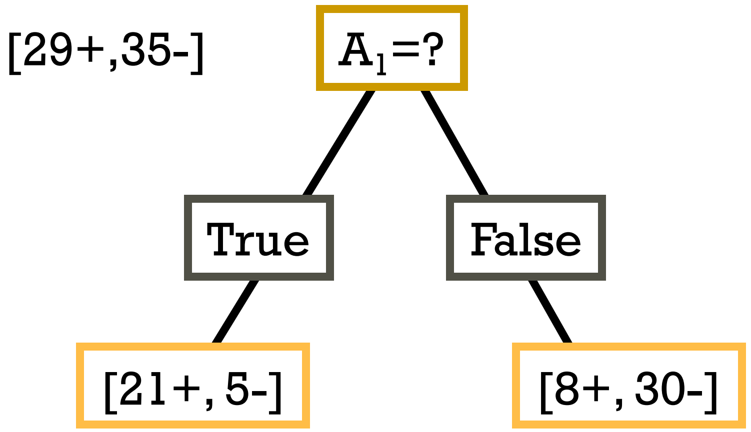


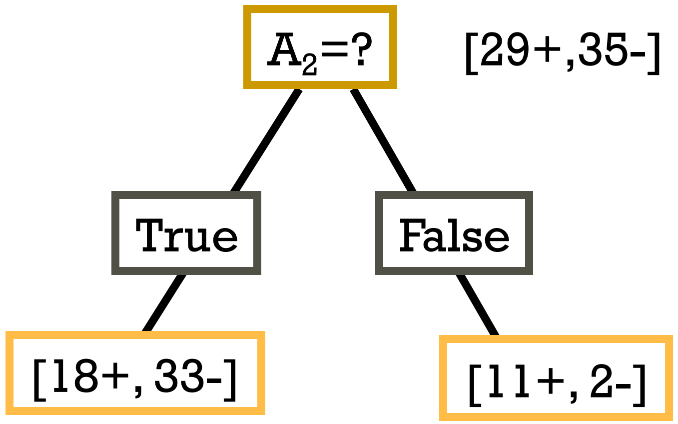
INFORMATION GAIN

$$\text{Gain}(S,A)=\text{Entropy}(S) - \sum_{v\in \text{values}(A)} |S_v|/|S| \ \text{Entropy}(S_v)$$

Entropy([21+,5-]) = 0.71
Entropy([8+,30-]) = 0.74
$\text{Gain}(S,A_1)$=Entropy(S)
-26/64*Entropy([21+,5-])
-38/64*Entropy([8+,30-])
=0.27

Entropy([18+,33-]) = 0.94
Entropy([8+,30-]) = 0.62
$\text{Gain}(S,A_2)$=Entropy(S)
-51/64*Entropy([18+,33-])
-13/64*Entropy([11+,2-])
=0.12



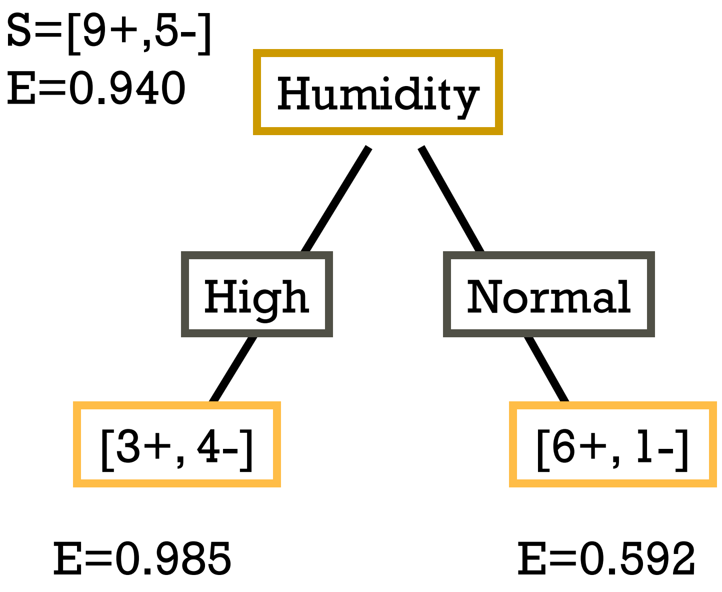


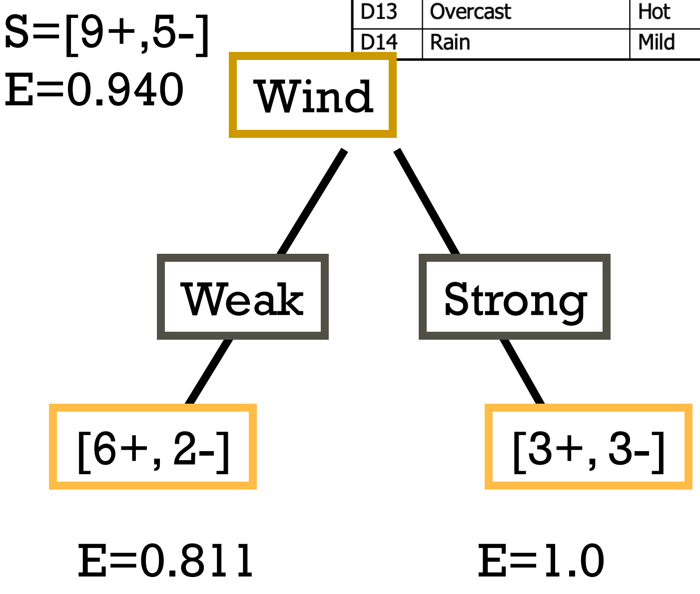
TRAINING EXAMPLES

Day	Outlook	Temp.	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cold	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

SELECTING THE NEXT ATTRIBUTE

Day	Outlook	Temp.	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cold	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No





Gain(S,Humidity)
=0.940-(7/14)*0.985
– (7/14)*0.592
=0.151

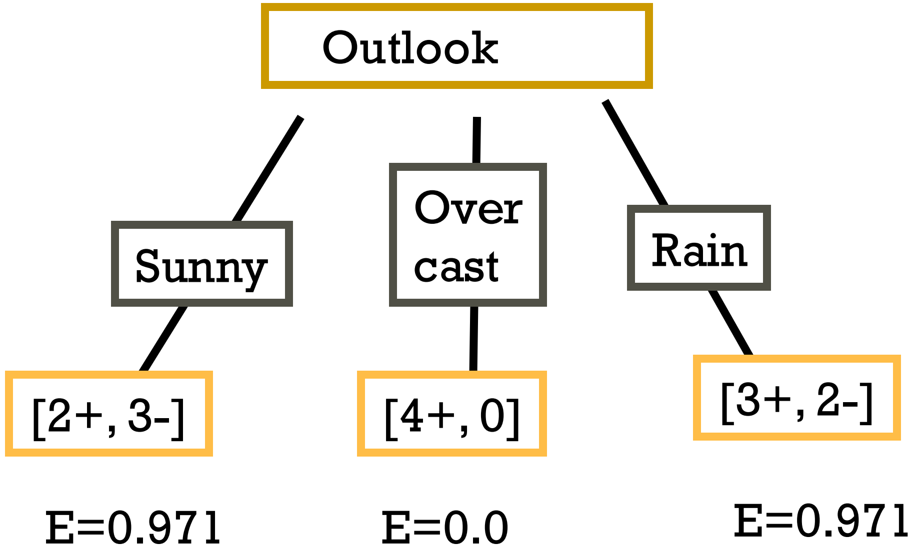
Gain(S,Wind)
=0.940-(8/14)*0.811
– (6/14)*1.0
=0.048

Humidity provides greater info. gain than Wind, w.r.t target classification.

Day	Outlook	Temp.	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cold	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

SELECTING THE NEXT ATTRIBUTE

S=[9+,5-]
E=0.940



Gain(S,Outlook)
=0.940-(5/14)*0.971
-(4/14)*0.0 – (5/14)*0.0971
=0.247

SELECTING THE NEXT ATTRIBUTE

- The information gain values for the 4 attributes are:
  - Gain(S,Outlook) =0.247
  - Gain(S,Humidity) =0.151
  - Gain(S,Wind) =0.048
  - Gain(S,Temperature) =0.029

- where S denotes the collection of training examples

Day	Outlook	Temp.	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cold	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

ID3 ALGORITHM

Day	Outlook	Temp.	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cold	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
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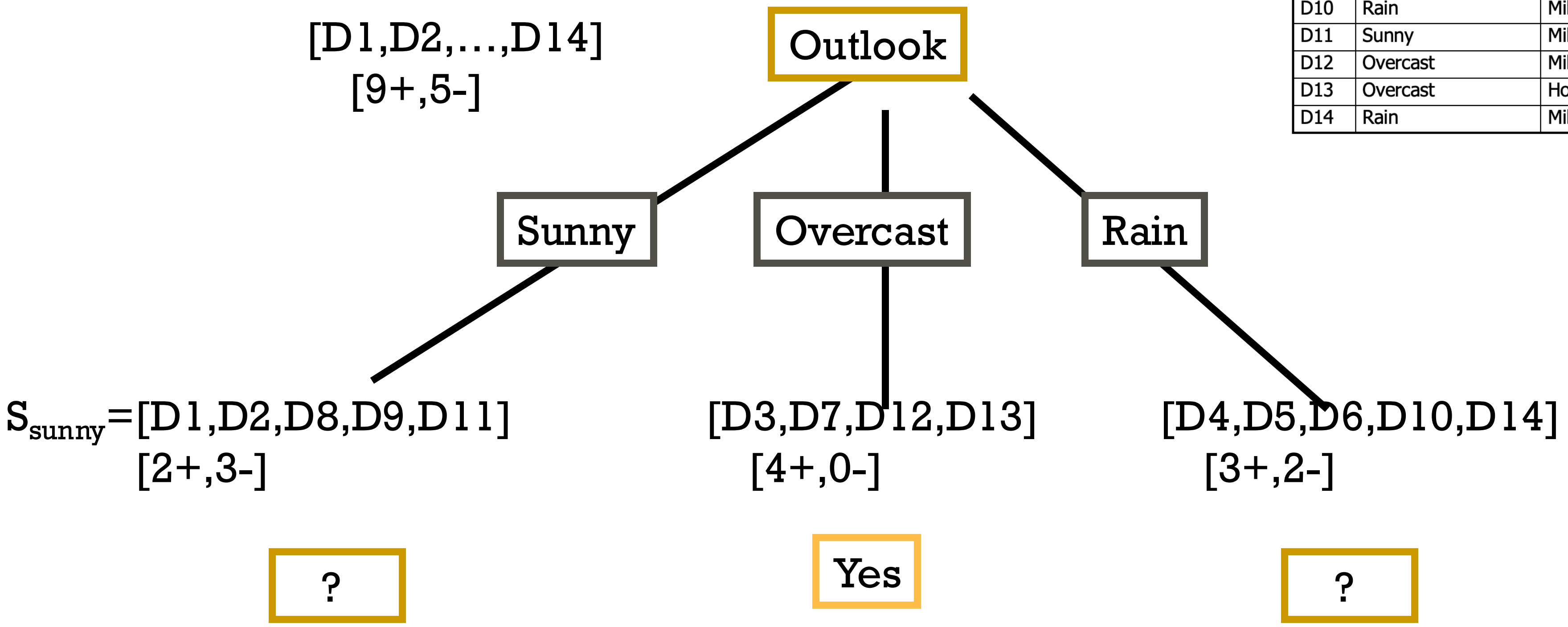
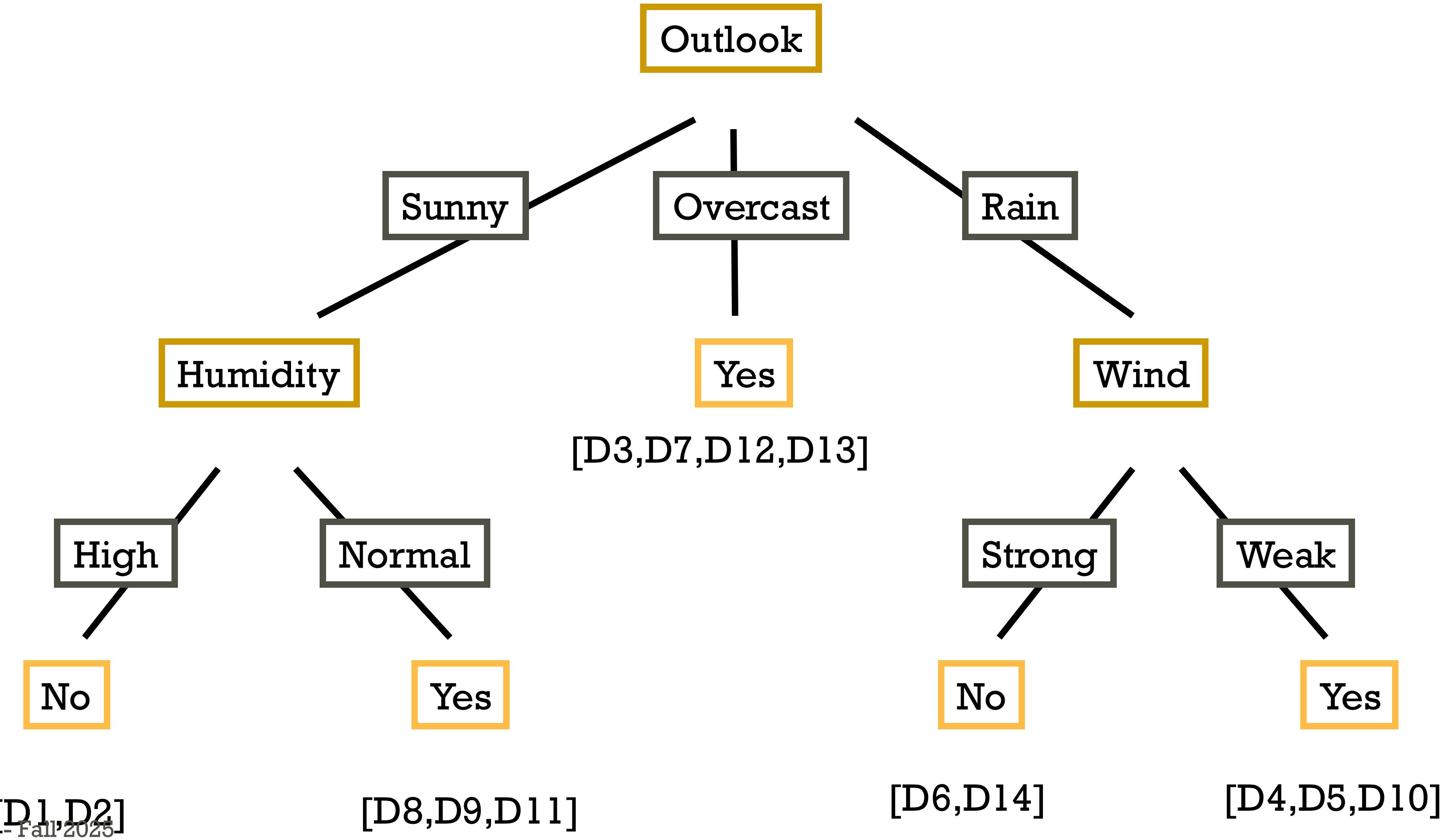


$\text{Gain}(S_{sunny}, \text{Humidity})=0.970-(3/5)0.0 - 2/5(0.0) = 0.970$

$\text{Gain}(S_{sunny}, \text{Temp.})=0.970-(2/5)0.0 - 2/5(1.0)-(1/5)0.0 = 0.570$

$\text{Gain}(S_{sunny}, \text{Wind})=0.970= -(2/5)1.0 - 3/5(0.918) = 0.019$

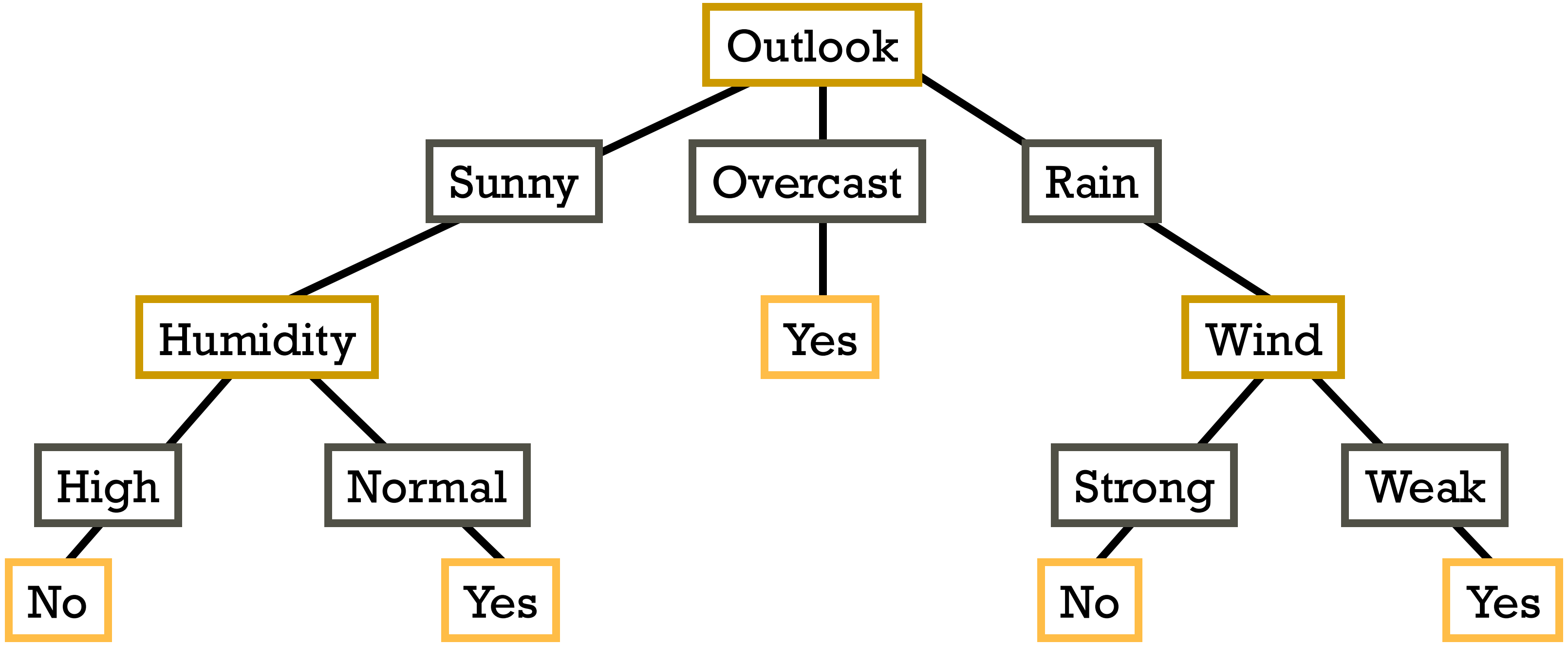
ID3 ALGORITHM



TOP-DOWN INDUCTION OF DECISION TREES ID3

1. A ← the "best" decision attribute for next *node*
2. Assign A as decision attribute for *node*
3. For each value of A create new descendant
4. Sort training examples to leaf node according to
   - the attribute value of the branch
5. If all training examples are perfectly classified (same value of target attribute) stop, else iterate over new leaf nodes.

CONVERTING A TREE TO RULES



R_1: If (Outlook=Sunny) $\wedge$ (Humidity=High) Then PlayTennis=No
R_2: If (Outlook=Sunny) $\wedge$ (Humidity=Normal) Then PlayTennis=Yes
R_3: If (Outlook=Overcast) Then PlayTennis=Yes
R_4: If (Outlook=Rain) $\wedge$ (Wind=Strong) Then PlayTennis=No
R_5: If (Outlook=Rain) $\wedge$ (Wind=Weak) Then PlayTennis=Yes

CONTINUOUS VALUED ATTRIBUTES

Create a discrete attribute to test continuous

- Temperature = 24.5^0C
- (Temperature > 20.0^0C) = {true, false}

Where to set the threshold?

Temperature	15^0C	18^0C	19^0C	22^0C	24^0C	27^0C
PlayTennis	No	No	Yes	Yes	Yes	No

(see paper by [Fayyad, Irani 1993]

Out of scope of this course

ATTRIBUTES WITH MANY VALUES

- **Problem:** if an attribute has many values, maximizing *InformationGain* will select it.
- E.g.: Imagine using Date=27.3.2002 as attribute perfectly splits the data into subsets of size 1

A Solution:

Use *GainRatio* instead of information gain as criteria:

GainRatio(S,A) = Gain(S,A) / SplitInformation(S,A)

SplitInformation(S,A) $= -\sum_{i=1..c} |S_i|/|S| \log_2 |S_i|/|S|$

Where S_i is the subset for which attribute A has the value v_i

Out of scope of this course

UNKNOWN ATTRIBUTE VALUES

What if some examples have missing values of A?

Use training example anyway sort through tree

- If node n tests A, assign most common value of A among other examples sorted to node n.
- Assign most common value of A among other examples with same target value
- Assign probability pi to each possible value vi of A
  - Assign fraction pi of example to each descendant in tree

Classify new examples in the same fashion

OVERFITTING

- One of the biggest problems with decision trees is **Overfitting**

OVERFITTING IN DECISION TREE LEARNING

- Can always classify training examples perfectly
  - keep splitting until each node contains 1 example
  - singleton = pure
- Doesn't work on new data

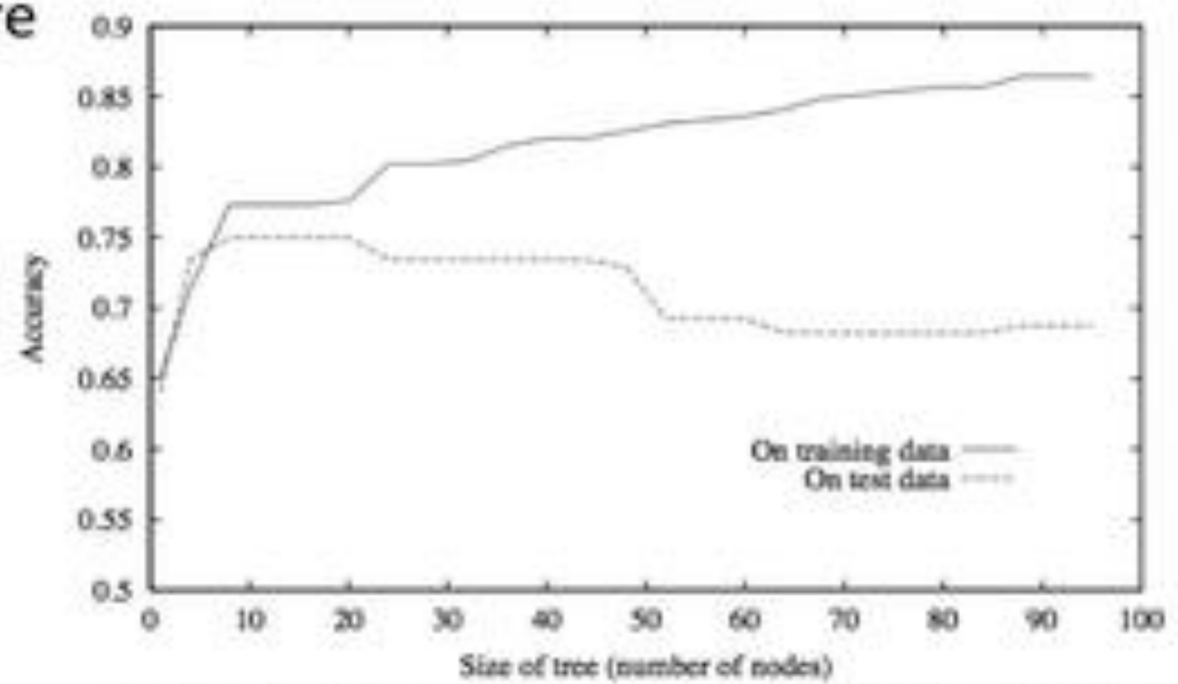


Figure credit: Tom Mitchell, 1997

WHEN TO CONSIDER DECISION TREES

- Instances describable by attribute-value pairs
- Target function is discrete valued
- Disjunctive hypothesis may be required
- Possibly noisy training data
- Missing attribute values
- Examples:
  - Medical diagnosis
  - Credit risk analysis
  - Object classification for robot manipulator (Tan 1993)

READING MATERIAL

- Chapter# 6: Principle of Data Science
- Chapter# 3, 6 & 7: Hands-On ML [Aurélien]